Date: 24 AUG 26
From: Midn 2/C Zakari
To: Prof O'Brien
Subj: **EW305 LAB 1 REPORT**
Ref: (a) Lab 1 handout
(b) Course notes
Encl: (1) List as needed (e.g., MATLAB scripts, .mat files)
(2) Must be referenced in the letter in the order listed here

1. Objective

The purpose of this Lab was to practice basic MATLAB skills i.e generating plots and labeling the plots and writing functions and checking the modeling accuracy by using NRMSE.

2. Results

This section contains your succinct explanations and lab deliverables (plots, equations, tables).

a. Calculated Values: My computed steady state value was 125.7864 for the data. I calculated the steady state Value by calculating the average of the final 100 data points to smooth out the noise.

b. NRMSE Calculation and Equation: My calculated NRMSE was 11.0226 percent, this shows that our error was little and most of our error was coming from the noise from the sensors rather than from experimental error.

c. **Graph of speed response NRMSE:**

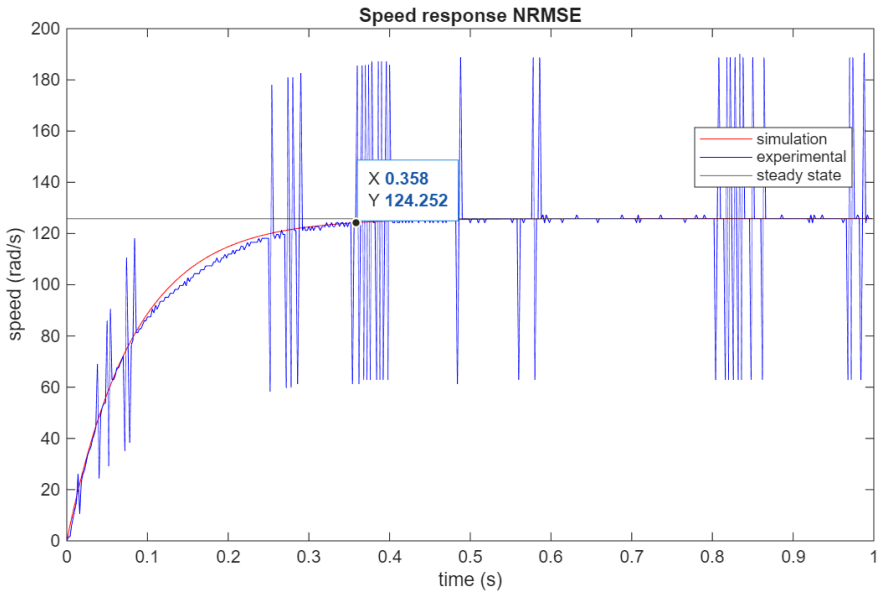


Figure 1: Experimental vs. Simulated DC Motor Speed Response

d. **Gen AI reflection:** Experimental systems usually have sensor noise, high frequencies and mechanical friction variations, whereas simulation models are ideal and do have no noise. Normalizing the root mean square error by dividing by the range (max - min) scaled the

error relative to the systems capacity and allowed us to have consistent accuracy comparisons across all the physical scales.

Midn 2/C Abdul-Majeed Zakari (signature)

References

[1] EW305 Instructor, *EW305 Lab Handouts*, Weapons, Robotics & Controls Engineering Department, USNA, AY27.